Data 780 Final Project Proposal

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Objective) Banks are incredibly vague and secretive with the criteria they use to judge people's applications for home loans. People rarely know which lenders are likely to say yes, or what the loan will cost, until after they apply. A denial costs time, fees, and credit. We want to build a free, open model, with a simple tool if time permits, that takes a person's situation and shows which North Carolina lenders are most likely to approve them, at what price, and why. We will also check whether the tool treats different groups of people fairly.

Task and goals) For each applicant profile and lender, predict the probability of approval and the expected loan cost, then rank lenders. Our goals are to produce accurate rankings, trustworthy probabilities, and recommendations that hold up across groups. We hope to learn which methods and data-preparation choices matter most. We want to know if our recommender can recover the best lender when applicants self-select, and whether removing demographic inputs changes outcomes.

Today) Online estimators are tied to a single lender or lead-generation site, and their criteria are not public. Applicants cannot compare lenders side by side or see why they were scored a certain way. These estimates may also overlook other factors that could raise or lower a borrower's chance of approval. Debt and credit scores change over time, so any estimate can go stale; we use recent data and design the method to be rerun on each new release. To our knowledge, there is no public validation of how accurate these estimates are. Lenders weigh credit score and debt-to-income, but public data has no credit scores and only ranges of debt-to-income, so our tool targets lender choice.

What's new) This model can integrate other factors such as location, and predict the likely interest rate, beyond the few factors currently used, to show a more accurate affordability for the borrower. It will also take into account available mortgage lenders, showing truly how affordable the mortgage will be for the borrower with specific lenders and which lender gives them the best potential of approval.

Existing mortgage estimators are proprietary, tied to a single lender, and, to our knowledge, unvalidated against any ground truth. We build an open lender-comparison model that estimates, for a given applicant profile, each lender's approval probability and expected price, and returns a ranking. Its novelty is a validation design: because public records show each applicant at only one lender, we first test the recommender on simulated markets where applicants self-select toward lenders and true lender quality is known, measuring how often each method recovers the true best lender. We then audit the recommender for fairness by comparing versions with and without demographic inputs, measuring whether the recommendations and their accuracy differ by group. We expect this to work because simulation lets us separate real lender quality from self-selection, which real data cannot.

Method) Our baseline method will utilize logistic regression to predict approval, and linear regression to predict price. More flexible methods include a decision tree and a random forest. The tree and forest can also predict price. Lenders are then ranked by predicted approval chance and price. We will interpret results by looking at which inputs matter most in the logistic regression, tree, and forest. To test extensions, we'll compare models with and without demographic inputs, and compare one model for all lenders against a separate model for each of the largest North Carolina lenders (a stretch goal, pursued only if the core work is on schedule). We will compare the models against each other, a few settings per model (for example how deep the tree is), and two data-preparation choices (for example, dropping vs. filling in missing values).

Data) The data comes from the Home Mortgage Disclosure Act (HMDA) Modified Loan/Application Register (LAR), restricted to North Carolina (NC), South Carolina, and Texas. Large files are sampled by state and year (five year sample of 2020-2024). We will also be creating synthetic data to simulate a market with lenders whose approval rules and prices are known, applicants who self-select toward lenders, and a deliberately built-in difference between groups (a planted disparity). We check whether each method recovers the true best lender and detects the planted disparity. The scope is home-purchase loans only. First-time-buyer programs, property taxes, insurance, and inflation are not modeled.

Evaluation) We have five evaluation metrics. Approval: ROC-AUC and precision-recall AUC (most applications are approved, so plain accuracy can be misleading). Probabilities: Brier score and calibration plots. Price: RMSE. Recommender (using simulated data): how often the true best lender appears in the top 3. Fairness: gaps in approval rates and error rates by group, with and without demographic inputs. Examples are chosen by random sampling or fixed profiles.

Who Cares) North Carolina homebuyers, especially lower-income or non-standard applicants, often apply blind. If we succeed, they get an open, cross-lender comparison built on public data instead of single-bank estimates. Comparing models with and without demographic inputs measures disparities in lending. Following FAIR practices (findable, accessible, interoperable, reusable) means every recommendation traces back to its data and methods.

Risks and Payoffs) Risks: Public HMDA lacks credit scores and reports debt-to-income in ranges, so apparent gaps may reflect unseen factors; we report them as disparities, and test sensitivity to the missing variables. Proxies like census tract and income can carry bias even without demographic fields; we compare models with and without tract and demographic inputs. Data lags behind lender policy, but the method can be rerun on each new annual release. Each applicant appears at only one lender; the simulation addresses this.

Payoffs: Fewer wasted applications and better lender matches for consumers, a reproducible measurement of lending fairness in NC, and a transparent model that contrasts with proprietary estimates.

Timeline) Week 1: extract and clean data, set scope filters. Weeks 2–3: baselines and data-preparation comparison on NC. Week 4: full model comparison across states and the price model. Week 5: simulator, recommender validation, fairness audit. Week 6: interpretation, best-practices list,

write-up, repository cleanup, and presentation. Per-lender models and the dashboard are stretch goals if we are ahead of schedule. Weekly team check-ins throughout. So far we have identified datasets, defined outcomes, agreed on scope, and explored the outcomes for a specific company, Truist, to see how their results compare to other major banks in NC.

Success) We succeed if our approval models clearly beat the logistic baseline on AUC or we show clearly when the simple baseline is good enough, and are well calibrated on held-out data and on the additional states. For the simulated markets, the recommender should find the true **best** lender more often than random ranking or the baseline. The fairness audit will produce clear group-level comparisons, reported as disparities, and finally, we will deliver an open repository and a best-practices list for HMDA-based lender modeling, backed by our results. A stretch goal would be creating a dashboard where users enter a profile and see ranked lenders with reasons.

Tools) Python (pandas, scikit-learn, matplotlib, additional python packages as need arises), public GitHub repository, optional Streamlit dashboard.